

The 35-Day Study Plan

Baseline schedule: approximately 5 hours/day of structured learning, plus project time. Do not treat every day as a one-topic lesson; several days deliberately combine related concepts.

Day	Learning focus	Practical output
Day 1 — ML Foundations + Mathematics	ML terminology, supervised vs unsupervised, parameters vs hyperparameters, vectors, matrices, dot products, derivatives, gradients, optimization.	Project 1: implement the mathematics of linear regression with NumPy.
Day 2 — Linear Regression	MSE, normal equation, gradient descent, coefficients, intercept, residuals, assumptions.	House-price regression; compare normal equation and gradient descent.
Day 3 — Regression Evaluation + Validation	MAE, MSE, RMSE, R^2 , adjusted R^2 , train/test split, validation, cross-validation.	Business revenue prediction + metric comparison.
Day 4 — Polynomial Regression + Regularization	Polynomial features, bias/variance, Ridge, Lasso, Elastic Net.	Demand prediction comparing linear, polynomial, Ridge, Lasso, Elastic Net.
Day 5 — Logistic Regression	Sigmoid, probability, odds/log-odds, log loss, decision boundary, gradient descent.	Customer churn prediction.
Day 6 — Classification Evaluation	Confusion matrix, precision, recall, F1, specificity, ROC-AUC, PR-AUC, threshold tuning.	Fraud detection; demonstrate why accuracy can be misleading.
Day 7 — Week 1 Combined	End-to-end supervised learning workflow.	Credit-risk prediction: preprocessing → baseline → model → validation → metrics → error analysis.
Day 8 — Decision Trees	Entropy, information gain, Gini impurity, splits, depth, pruning, overfitting.	Customer classification.
Day 9 — Random Forest	Bagging, bootstrap sampling, randomized features, OOB evaluation, feature importance.	Network intrusion classification.
Day 10 — KNN + SVM	Distance metrics, K, scaling, hyperplanes, margins, support vectors, C, kernels.	Security-event classification; compare multiple models.
Day 11 — Naive Bayes + Classical NLP	Bayes theorem, conditional independence, Bag of Words, TF-IDF, n-grams.	Spam/phishing text classifier.
Day 12 — Gradient Boosting	Weak learners, residual correction, loss minimization, learning rate, estimator count.	Financial-risk prediction.
Day 13 — XGBoost / LightGBM / CatBoost	Practical gradient-boosting families, hyperparameters, categorical handling, regularization.	Serious tabular model comparison.

Day 14 — Week 2 Combined	Model comparison and cybersecurity workflow.	Cybersecurity threat detection: baseline → RF → boosting → tuning → error analysis.
Day 15 — K-Means	Centroids, assignment/update loop, inertia, elbow method, initialization.	Customer segmentation.
Day 16 — Hierarchical Clustering	Agglomerative clustering, linkage, dendrograms.	Business customer segmentation.
Day 17 — PCA	Covariance, eigenvectors/eigenvalues, explained variance, projection.	Dimensionality reduction + downstream model.
Day 18 — Anomaly Detection	Isolation Forest, One-Class SVM, LOF, anomaly scores.	Cybersecurity anomaly detection.
Day 19 — Feature Engineering	Interactions, ratios, aggregations, temporal and domain features, transformations.	Improve a weak model without changing the algorithm.
Day 20 — Feature Selection + Explainability	Mutual information, chi-square, RFE, Lasso, permutation importance, SHAP concepts.	Explain model decisions and reduce unnecessary features.
Day 21 — Week 3 Combined	Supervised + unsupervised workflow.	Financial anomaly/risk detection combining feature engineering, PCA, clustering, anomaly detection, and supervised ML.
Day 22 — Validation Architecture	Train/validation/test, K-fold, stratification, time splits, leakage, baselines.	Intentionally introduce and then fix leakage in a previous project.
Day 23 — Hyperparameter Optimization	Grid search, randomized search, successive-halving concepts, Bayesian optimization concept.	Tune the best model from an earlier project.
Day 24 — sklearn Pipelines	Pipeline, ColumnTransformer, transformation/model composition.	Convert an old notebook into a reproducible training pipeline.
Day 25 — ML Project Architecture	Project structure, modules, configuration, logging, testing, separation of training and inference.	Refactor a project into an engineering-grade repository.
Day 26 — Experiment Tracking	MLflow experiments, runs, parameters, metrics, artifacts, model logging.	Track multiple model experiments.
Day 27 — Model Registry + Lifecycle	Model versions, lineage, metadata, aliases, approval and rollback concepts.	Register and version the best model.
Day 28 — Model Serving	Serialization, model loading, FastAPI, request validation, inference.	Create /predict, /health, and /model-info.
Day 29 — Complete ML API	API testing, errors, logging, latency, schema stability.	Turn a trained pipeline into a usable inference service.
Day 30 — 30-Day ML System Checkpoint	End-to-end system architecture.	Training → validation → MLflow → model artifact → FastAPI.
Day 31 — Docker Fundamentals	Image, container, Dockerfile, ports, environment variables, .dockerignore.	Containerize the FastAPI ML service.

35-Day Applied Machine Learning & ML Engineering Roadmap

Day 32 — Docker + ML	Production-oriented container configuration, health checks, runtime dependencies.	Run and test the complete ML container.
Day 33 — Cloud Fundamentals	Compute, storage, networking, secrets, logs, service exposure.	Choose a simple deployment target and prepare the service.
Day 34 — Cloud Deployment	Deploying a containerized FastAPI ML service.	Put one ML API online and test it.
Day 35 — Final ML Engineer Project	Integrate algorithms, evaluation, pipelines, tracking, serving, Docker and cloud.	Finish one flagship system and polish the portfolio.